

v2.0

Stripe Billing Service

Complete Feature Documentation

Multi-tenant SaaS billing platform with Stripe integration

Platform	React + ASP.NET Core
Architecture	Multi-Tenant SaaS
Integration	Stripe API
Version	2.0 — March 2026

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1. Platform Overview

The Stripe Billing Service is a comprehensive, multi-tenant SaaS billing platform built with a React (TypeScript) frontend and an ASP.NET Core backend. It integrates deeply with the Stripe API to provide end-to-end billing operations including payment processing, subscription management, invoicing, refunds, dunning, usage-based billing, and more.

The platform follows a multi-tenant architecture where each tenant operates in complete data isolation. A SuperAdmin has platform-level oversight with the ability to create tenants, assign initial administrators, and monitor system-wide metrics without accessing individual tenant data directly.

Key Capabilities

- **Multi-Tenant Architecture** — Complete data isolation per tenant with tenant-scoped API access
- **Stripe Integration** — Native integration with Stripe for payments, subscriptions, invoices, and Connect
- **Role-Based Access Control** — Four-tier role hierarchy (SuperAdmin, Admin, Manager, Viewer)
- **Real-Time Analytics** — MRR tracking, churn analysis, revenue trends, and LTV calculations
- **API Gateway** — Managed API keys, request logging, webhook management, and event processing
- **Dunning & Recovery** — Automated failed payment recovery with configurable retry schedules
- **Usage-Based Billing** — Metered billing support with usage tracking and consumption analytics
- **Export & Reporting** — CSV and PDF exports for transactions, invoices, customers, and audit logs
- **Impersonation Mode** — SuperAdmin can view tenant data in read-only mode for support and debugging

Technology Stack

Layer	Technology	Details
Frontend	React 18 + TypeScript	Vite build system, React Router, Bootstrap 5, Recharts
Backend	ASP.NET Core 8	RESTful API, Entity Framework Core, JWT authentication
Database	SQL Server	Multi-tenant data isolation via TenantId column
Payments	Stripe API	Payments, Subscriptions, Invoices, Connect, Tax
Auth	JWT + BCrypt	Access tokens, refresh tokens, role-based policies

2. User Roles & Authorization

The platform implements a strict role-based access control (RBAC) system with four distinct roles. Each role has specific permissions that determine which features and data a user can access.

2.1 Role Hierarchy

Role	Level	Scope	Description
SuperAdmin	100	Platform	Platform administrator. Manages tenants, creates initial admins, monitors system-wide metrics. Cannot access tenant billing data directly.
Admin	50	Tenant	Tenant administrator. Full access to all billing features, user management, settings, and configuration within their tenant.
Manager	20	Tenant	Operational role. Can view all billing data, run analytics, and manage day-to-day billing operations. Cannot manage users or settings.
Viewer	10	Tenant	Read-only access. Can view dashboards, payments, subscriptions, and reports. Cannot create, modify, or delete any records.

2.2 SuperAdmin Capabilities

The SuperAdmin role is fundamentally different from tenant roles. SuperAdmins operate at the platform level and are intentionally excluded from tenant-scoped authorization policies. This ensures that platform administrators cannot inadvertently access or modify individual tenant data.

SuperAdmin-Exclusive Features:

- **Tenant Management** — Create, update, suspend, and activate tenants
- **Admin User Creation** — Create the first admin user for each new tenant
- **Platform Dashboard** — View system-wide metrics across all tenants
- **Platform Analytics** — Revenue breakdown and performance by tenant
- **Global Email Templates** — Manage platform-wide email notification templates
- **Platform Settings** — Configure platform name, currency, maintenance mode
- **Impersonation** — View tenant data in read-only mode for support

2.3 Impersonation Mode

Impersonation allows a SuperAdmin to view a specific tenant's billing data without having direct access to tenant-scoped endpoints. When impersonating, the system generates a short-lived (30-minute) JWT token with a Viewer role scoped to the target tenant. A prominent yellow banner is displayed at the top of the screen indicating read-only mode, and all non-GET API requests are blocked at the client level.

Security Note: Impersonation tokens expire after 30 minutes, carry the "IsImpersonating" claim, and the original SuperAdmin token is preserved for seamless session restoration. All write operations are blocked during impersonation.

3. Dashboard

3.1 Tenant Dashboard

The tenant dashboard provides an at-a-glance overview of key billing metrics and recent activity for the currently logged-in tenant. It is the default landing page for Admin, Manager, and Viewer users.

Metrics Cards

Metric	Description	Data Source
Total Revenue	Cumulative revenue from all successful payments	Aggregated from Payment records
MRR	Monthly Recurring Revenue calculated from active subscriptions	Sum of active subscription amounts
Active Subscriptions	Count of subscriptions in 'active' or 'trialing' status	Subscription records
Total Customers	Total number of registered customers for the tenant	Customer records count

Each metric card includes a trend indicator showing the change direction compared to the previous period.

30-Day Revenue Trend Chart

A line chart displaying daily revenue over the past 30 days. The x-axis represents dates and the y-axis shows revenue amounts. This visualization helps identify revenue patterns, seasonal trends, and anomalies.

Recent Activity Feed

Column	Description
Type	The category of activity (payment, subscription change, refund, etc.)
Description	Human-readable description of the event
Amount	Monetary value associated with the activity, formatted in tenant currency
Status	Current status shown as a color-coded badge (succeeded, pending, failed)
Time	Timestamp of when the activity occurred, shown in relative format

3.2 SuperAdmin Dashboard

The SuperAdmin dashboard provides platform-wide metrics aggregated across all tenants. This is the default landing page for SuperAdmin users and is only accessible to the SuperAdmin role.

Platform Metrics

Metric	Description
Total Tenants	Total number of tenant organizations registered on the platform
Active Tenants	Count of tenants with isActive = true (not suspended)
Total Customers	Sum of all customers across all tenants
Active Subscriptions	Sum of all active subscriptions across all tenants
Total Revenue	Combined revenue from all tenants, formatted as currency
Failed Payments (30d)	Count of failed payment attempts in the last 30 days across all tenants

Below the metrics, a table shows the most recently created tenants with their name, active status, and creation date.

4. Billing Module

The billing module is the core of the platform, providing comprehensive management of payments, subscriptions, customers, invoices, refunds, and more. All billing data is tenant-scoped.

4.1 Payments

The Payments page displays all payment transactions for the current tenant. Payments are synced from Stripe and represent completed, pending, or failed charge attempts against customer payment methods.

Column	Description
Customer Name	Full name of the customer who made the payment
Amount	Payment amount formatted in the tenant's currency (e.g., \$120.00)
Status	Badge indicating: Succeeded (green), Pending (yellow), Failed (red)
Payment Method	Card brand and last 4 digits (e.g., 'Visa **** 4242')
Date	Date and time when the payment was processed

Filters: Search by customer name or email. Filter by status (All, Succeeded, Pending, Failed).

Actions: Export button to download payment data.

4.2 Subscriptions

Manages recurring billing subscriptions. Each subscription links a customer to a pricing plan and tracks the billing cycle, renewal date, and current status.

Column	Description
Customer Name	The subscriber's full name
Plan	Name of the subscription plan (e.g., 'Professional Monthly')
Amount	Recurring charge amount per billing period
Status	Active (green), Trialing (blue), Past Due (orange), Cancelled (gray)
Renews	Next renewal date for the subscription billing cycle

Filters: Search by customer name. Filter by status (Active, Trialing, Past Due, Cancelled).

Actions: 'New Subscription' button to create a subscription.

4.3 Customers

The customer registry for the current tenant. Each customer record is synced with Stripe and contains contact information, subscription counts, and lifetime spending.

Column	Description
Name	Customer's full name as stored in Stripe
Email	Primary email address used for billing communications
Subscriptions	Count of active subscriptions displayed as a numeric badge
Total Spent	Lifetime total of all successful payments by this customer

Filters: Search by name or email.

Actions: 'Add Customer' button to register a new customer.

4.4 Invoices

Invoice management with full lifecycle tracking from draft to paid. Invoices can be generated automatically from subscriptions or created manually. Each invoice supports PDF download.

Column	Description
Invoice #	Unique invoice number assigned by Stripe (e.g., 'INV-0042')
Customer	Customer name associated with the invoice
Amount	Total invoice amount including line items and tax
Status	Paid (green), Draft (gray), Open (blue) — indicates payment state
Date	Invoice creation or issue date
Action	Download PDF button to retrieve the invoice document from Stripe

Filters: Search by invoice number. Filter by status (Paid, Draft, Open).

Actions: 'Create Invoice' button. Click any invoice to open detail viewer with PDF preview.

4.5 Refunds

Tracks all refund requests and their processing status. Refunds are processed through Stripe and can be full or partial. Each refund is linked to the original payment.

Column	Description
Customer Name	Name of the customer receiving the refund
Amount	Refund amount (may be less than original payment for partial refunds)
Reason	Reason for the refund (e.g., 'Duplicate charge', 'Product unsatisfactory')
Status	Pending (yellow), Processing (blue), Completed (green), Failed (red)
Requested	Date when the refund was initially requested

Actions: 'Request Refund' button to initiate a new refund.

4.6 Coupons & Promotions

Two-tab interface for managing discount coupons and their associated promotion codes. Coupons define discount rules while promotion codes are customer-facing codes that apply coupons.

Coupons Tab

Column	Description
Name	Descriptive name for the coupon (e.g., 'Summer Sale 20%')

Column	Description
Discount	Percentage or fixed amount badge (e.g., '20%' or '\$10.00')
Duration	How long the discount applies: once, repeating, or forever
Redeemed	Number of times this coupon has been used
Status	Active (green) or Inactive (gray)
Actions	Create Promo Code, Deactivate buttons

Promotion Codes Tab

Column	Description
Code	The customer-facing alphanumeric code (e.g., 'SAVE20')
Coupon Name	The parent coupon this promotion code applies
Redeemed	Number of times this specific code has been used
Status	Active or Inactive
Expires	Expiration date for the promotion code, if set

Create Coupon Form: Name, Type (percent/fixed), Percentage or Amount, Duration, Max Redemptions.

Create Promo Code Form: Code string, Max Redemptions, First-time customers only toggle.

4.7 Plans

Displays available subscription plans in a visual grid layout using Plan Cards. Each plan card shows the plan name, pricing, billing interval, and included features. Plans are synced from Stripe Products and Prices.

Actions: 'Create Plan' button to define a new subscription plan with pricing and features.

4.8 Usage Billing

Metered/usage-based billing management. This page tracks consumption-based charges where customers are billed based on actual usage rather than flat subscription fees.

Metrics Cards

Metric	Description
Current Period Usage	Total usage units consumed in the current billing period
Estimated Revenue	Projected revenue based on current usage and per-unit pricing
Active Meters	Number of active usage meters tracking consumption
Top Consumers	Count of highest-usage customers in the current period

Usage Trend Chart

Bar chart showing daily usage volume. Helps identify consumption patterns and peak usage periods.

Recent Usage Records Table

Column	Description
Subscription ID	Truncated ID linking usage to a specific subscription
Quantity	Number of units consumed in this record
Action	Increment or decrement badge showing the type of usage adjustment
Timestamp	Exact date and time the usage was recorded
Idempotency Key	Unique key to prevent duplicate usage records

4.9 Credits & Balances

Manages customer credit balances. Credits can be issued as goodwill gestures, promotional offers, or billing adjustments. Positive credits reduce future invoice amounts.

Metrics Cards

Metric	Description
Total Outstanding Credits	Sum of all unused credit balances across customers
Total Credits Issued	Cumulative value of all credits ever issued
Customers with Balance	Count of customers who currently have a non-zero credit balance

Credit Transaction Table

Column	Description
Customer ID	Truncated Stripe customer identifier
Amount	Credit or debit amount with +/- indicator and color coding
Type	Credit (green badge) or Debit (red badge)
Description	Reason for the credit adjustment
Balance After	Customer's remaining credit balance after this transaction
Date	Date and time the credit was applied

Add Credit Form: Customer ID, Amount (in cents), Currency, Description.

4.10 Dunning Management

Automated failed payment recovery system. When a payment fails, the dunning system creates a retry schedule to attempt recovery. Administrators can pause, resume, or cancel dunning schedules.

Metrics Cards

Metric	Description
Active Schedules	Number of dunning schedules currently running retry attempts
Recovered	Count of invoices successfully recovered through the dunning process
Failed	Count of invoices that exhausted all retry attempts without recovery
Recovery Rate	Percentage of dunning cases that resulted in successful payment recovery

Dunning Schedules Table

Column	Description
Invoice ID	Truncated Stripe invoice identifier linked to the failed payment
Status	Active (blue), Paused (yellow), Completed (green), Cancelled (gray)
Step	Current retry step out of maximum (e.g., '2/4')
Next Retry	Scheduled date and time for the next payment retry attempt
Retries	Total number of retry attempts made so far
Started	Date when the original payment failure triggered the dunning schedule
Actions	Pause, Resume, or Cancel buttons depending on current status

5. Analytics

5.1 Tenant Revenue Analytics

Comprehensive revenue and subscription analytics for the current tenant. Provides insights into monthly recurring revenue, customer lifetime value, churn rates, and revenue growth trends.

Key Metrics

Metric	Description	Calculation
Current MRR	Monthly Recurring Revenue	Sum of all active subscription amounts normalized to monthly
Monthly Churn Rate	Percentage of subscriptions lost	Churned subscriptions / Total subscriptions at period start
Average LTV	Customer Lifetime Value	Average revenue per customer over their entire relationship
Retention Rate	Customer retention percentage	100% minus churn rate

MRR Components

Component	Description
New MRR	Revenue from newly acquired subscriptions in the current period
Expansion MRR	Additional revenue from plan upgrades or add-ons
Contraction MRR	Revenue lost from plan downgrades
Churned MRR	Revenue lost from cancelled subscriptions

Customer Metrics

Metric	Description
Avg Subscription Duration	Average length of time customers maintain their subscriptions
Avg Revenue Per Customer	Mean revenue generated per customer across the tenant
Annual Churn Rate	Annualized percentage of customer/subscription churn
Churned Subscriptions	Total count of subscriptions cancelled in the analysis period

5.2 Platform Analytics (SuperAdmin)

System-wide analytics available only to SuperAdmin users. Shows aggregated metrics across all tenants with a revenue breakdown table showing per-tenant performance.

Column	Description
Tenant	Tenant organization name
Revenue	Total revenue generated by the tenant
Active Subscriptions	Count of active subscriptions within the tenant
Customers	Total number of customers registered under the tenant

6. API Gateway

The API Gateway module provides tools for managing API access, monitoring request activity, and handling Stripe webhook integrations.

6.1 API Keys

Manage API keys used for programmatic access to the billing platform. Keys can be scoped to different environments (production, sandbox) and tracked for usage.

Column	Description
Name	Descriptive label for the API key (e.g., 'Production Backend')
Environment	Target environment: Production or Sandbox
Status	Active (green) or Revoked (red) — indicates if the key is currently valid
Last Used	Timestamp of the most recent API call made with this key
Requests	Total number of API requests made using this key

Actions: 'Create API Key' button. Keys are shown once upon creation and cannot be retrieved later.

6.2 Request Logs

Real-time log of all API requests hitting the billing platform. Useful for debugging integration issues, monitoring API usage patterns, and identifying errors.

Column	Description
Endpoint	The API endpoint path that was called (e.g., '/v1/payments')
Method	HTTP method used: GET, POST, PUT, DELETE
Status	Success (green) or Error (red) badge based on response
HTTP	HTTP status code badge (200, 201, 400, 401, 404, 500, etc.)
Duration	Request processing time in milliseconds
Time	Timestamp when the request was received

Filters: Search by endpoint path. Filter by status (Success, Error).

6.3 Webhooks

Configure webhook endpoints to receive real-time notifications when events occur in the billing system. Webhooks push event data to external URLs for integration with other systems.

Column	Description
Description	Human-readable label for the webhook (e.g., 'CRM Integration')
URL	The endpoint URL where events will be delivered (shown in code format)
Status	Active (green) or Disabled (gray)
Created	Date when the webhook endpoint was registered

Actions: 'Add Webhook' button to register a new endpoint.

6.4 Webhook Events

Tracks all incoming Stripe webhook events and their processing status. Events that fail processing can be retried manually. Full event payloads are available for inspection.

Column	Description
Event Type	Stripe event type (e.g., 'invoice.paid', 'charge.succeeded') shown as a code badge
Stripe Event ID	Truncated unique identifier from Stripe (e.g., 'evt_1234...')
Status	Processed (green check), Failed (red X), Pending (yellow clock)
Retries	Number of processing attempts made for this event
Received	Timestamp when the event was received from Stripe
Actions	View (shows full JSON payload) and Retry (re-processes failed events) buttons

Filters: Search by event type. Filter by status. Manual refresh button.

7. Administration

Administration features are available to users with the Admin role within their tenant. These tools manage users, configure settings, review audit trails, and handle integrations.

7.1 User Management

Manage team members within the tenant. Admins can invite users, assign roles, and monitor login activity. The first admin for each tenant is created by the SuperAdmin.

Column	Description
Name	User's full name
Email	Login email address
Role	Assigned role badge: Admin (purple), Manager (blue), Viewer (gray)
Status	Active (green) or Inactive (red) — determines login ability
Last Login	Most recent successful login timestamp

Actions: 'Add User' button. Search by name or email.

7.2 Settings

Tenant-level configuration organized into four tabs:

Tab	Fields	Description
General	Company Name, Email	Basic tenant organization information
Billing	Billing preferences	Configure billing behavior and defaults
Notifications	Email notifications, Payment alerts	Toggle notification channels on/off
Security	Change Password	Account security and password management

7.3 Audit Logs

Complete audit trail of all actions performed within the tenant. Every create, update, and delete operation is logged with the acting user, timestamp, and affected entity.

Column	Description
User	Email of the user who performed the action

Column	Description
Action	Type of operation: Create, Update, or Delete
Entity Type	The type of record affected (Customer, Subscription, Payment, etc.)
Tenant	Tenant context in which the action occurred
Timestamp	Exact date and time of the action

Filters: Search by user email. Filter by action type (Create, Update, Delete).

7.4 Tax Configuration

Configure automatic tax calculation for invoices and subscriptions. Supports integration with Stripe Tax, TaxJar, or Avalara. Tax registrations can be added per jurisdiction.

Status Cards

Card	Description
Provider	Currently configured tax provider (Stripe Tax, TaxJar, Avalara)
Status	Whether tax calculation is currently active or inactive
Tax Behavior	Exclusive (added on top) or Inclusive (included in price)
Fallback Tax Rate	Default percentage applied when provider cannot determine exact rate

Configuration Form: Provider selection, auto-calculate toggle, default tax behavior, fallback rate, tax registrations by country.

7.5 Email Templates

Manage tenant-level email notification templates. Templates define the content sent for different billing events. The page has two tabs: Templates and Email Logs.

Templates Tab

Column	Description
Template	Template name and subject line displayed together
Status	Active or Inactive — determines if emails are sent for this event
Last Updated	Most recent modification date
Actions	Preview (renders HTML) and Edit buttons

Email Logs Tab

Column	Description
Recipient	Email address the notification was sent to
Subject	Email subject line
Status	Sent (green), Failed (red), or Pending (yellow)
Sent At	Timestamp when the email was dispatched

Create Template Form: Template Name, Category, Subject Line, HTML Body, Plain Text Body.

7.6 Export Center

Centralized data export functionality. Export billing data in CSV or PDF format with date range filtering. Six export categories are available.

Export Type	Description	Included Data
Transactions	Payment transaction records	Amount, status, customer, date, payment method
Invoices	Invoice records	Invoice number, amount, status, customer, dates
Customers	Customer data	Name, email, subscription count, total spent
Subscriptions	Subscription records	Plan, amount, status, customer, renewal dates
Refunds	Refund records	Amount, reason, status, customer, dates
Audit Logs	Audit trail data	User, action, entity type, timestamp

Settings: Format (CSV or PDF), From Date, To Date.

7.7 Stripe Connect

Manage Stripe Connect integrations for marketplace or platform business models. Connect allows the tenant to onboard sub-merchants and facilitate transfers.

Metrics Cards

Metric	Description
Connected Accounts	Number of Stripe Connect accounts onboarded
Total Transfers	Count of all transfers made to connected accounts
Transfer Volume	Total monetary value of all transfers

Connected Accounts Table

Column	Description
Account	Email and Stripe Account ID for the connected merchant
Type	Account type badge (Individual or Company)
Country	Two-letter country code of the connected account
Charges	Whether the account can accept charges (Enabled/Disabled)
Payouts	Whether the account can receive payouts (Enabled/Disabled)
Actions	Transfer button to send funds to this account

Add Account Form: Email, Business Type (Individual/Company), Country.

Create Transfer Form: Amount (in cents), Currency, Description.

8. SuperAdmin Module

The SuperAdmin module provides platform-level administration tools. These pages are only accessible to users with the SuperAdmin role and are served under the `/super-admin/*` route prefix. SuperAdmin users see a completely different sidebar navigation focused on platform management.

8.1 Tenant Management

Central hub for managing all tenant organizations on the platform. SuperAdmins can create new tenants, assign initial administrators, monitor tenant health, and use impersonation for support.

Tenants Table

Column	Description
Name	Tenant organization name
Revenue	Total revenue generated by the tenant, formatted as currency
Active Subs	Number of active subscriptions within the tenant (info badge)
Customers	Total customer count for the tenant (secondary badge)
Status	Active (green) or Inactive (red) badge
Actions	Two buttons: 'View as Tenant' (impersonate) and 'Create Admin'

Create Tenant Form

Field	Description
Tenant Name	Display name for the organization (e.g., 'Acme Corp')
Slug	URL-friendly identifier (e.g., 'acme-corp'), used in API routing
Description	Optional description of the tenant organization
Plan	Subscription tier: Free, Starter, Professional, or Enterprise

Create Admin Form

Creates the first administrator for a tenant. This action is only allowed when no admin user exists for the tenant. After the first admin is created, user management is delegated to the tenant admin.

Field	Description
Email	Login email for the new admin user (must be globally unique)
Password	Initial password (minimum 6 characters)

Field	Description
First Name	Optional first name
Last Name	Optional last name

8.2 Global Email Templates

Platform-wide email templates that serve as defaults for all tenants. These templates are stored with TenantId = empty (platform scope) and define the baseline email content for billing events.

Column	Description
Name	Template name (e.g., 'Payment Receipt')
Subject	Email subject line template
Event Type	Billing event that triggers this email (e.g., 'invoice.paid')
Status	Active (green) or Inactive (gray)
Actions	Edit and Delete buttons

Supported Event Types: invoice.created, invoice.paid, invoice.overdue, subscription.created, subscription.canceled, payment.failed, payment.succeeded, customer.created.

8.3 Platform Settings

Global platform configuration that applies to all tenants. Settings are persisted and can be updated at any time by the SuperAdmin.

Setting	Type	Description
Platform Name	Text	Name of the billing platform displayed in headers and emails
Default Currency	Select	Default currency for new tenants (USD, EUR, GBP, CAD, AUD)
Max Tenants Allowed	Number	Maximum number of tenants the platform can host
Default Features	JSON	Default feature flags for new tenants (JSON array format)
Maintenance Mode	Toggle	When enabled, tenant users see a maintenance message

End of Documentation

This document covers all features of the Stripe Billing Service platform as of version 2.0. For technical API documentation, integration guides, or developer resources, please contact the platform administrator.

Need Help? For questions about the platform, contact your tenant administrator. SuperAdmins can reach the platform team for infrastructure and configuration support.